

SYSC 5108 Exam Study

Assignment 3, Question 2

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Question

The artificial neuron takes three inputs. For the input vector

$$\mathbf{d} = [0.2, 0.5, 0.7],$$

calculate:

- (a) the weighted sum for the neuron,
- (b) the output if the activation function φ is a threshold activation with threshold $\theta = 1$,
- (c) the output if φ is the logistic function,
- (d) the output if φ is the rectified linear unit (ReLU) function.

Course and Textbook Cross-References

- **Chapter 4 slides, General Neural Network:** a neuron computes an affine or weighted sum first, then applies a nonlinear activation function.
- **Chapter 4 slides, Hidden Units:** ReLU is defined as

$$g(z) = \max(0, z).$$

- **Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 3:** the logistic sigmoid is

$$\sigma(x) = \frac{1}{1 + \exp(-x)}.$$

- **Goodfellow et al., Chapter 6:** neural-network hidden units usually compute an affine transformation and then apply an activation function; ReLU is defined as $g(z) = \max(0, z)$.

Given Values From the Diagram

The bias input is

$$d_0 = 1.$$

The weights are:

$$w_0 = 0.20, \quad w_1 = -0.10, \quad w_2 = 0.15, \quad w_3 = 0.05.$$

The input vector gives

$$d_1 = 0.2, \quad d_2 = 0.5, \quad d_3 = 0.7.$$

Step-by-Step Solution

Part (a): Weighted sum

The weighted sum is

$$z = w_0d_0 + w_1d_1 + w_2d_2 + w_3d_3.$$

Substitute the values:

$$\begin{aligned} z &= (0.20)(1) + (-0.10)(0.2) + (0.15)(0.5) + (0.05)(0.7) \\ &= 0.20 - 0.020 + 0.075 + 0.035 \\ &= 0.290. \end{aligned}$$

Therefore,

$$\boxed{z = 0.29}.$$

Part (b): Threshold activation with $\theta = 1$

Using a threshold activation,

$$\varphi(z) = \begin{cases} 1, & z \geq \theta, \\ 0, & z < \theta. \end{cases}$$

Here,

$$z = 0.29 < 1.$$

Therefore,

$$\boxed{\varphi(z) = 0}.$$

Part (c): Logistic activation

The logistic activation is

$$\varphi(z) = \sigma(z) = \frac{1}{1 + \exp(-z)}.$$

Using $z = 0.29$:

$$\begin{aligned} \sigma(0.29) &= \frac{1}{1 + \exp(-0.29)} \\ &\approx 0.5720. \end{aligned}$$

Therefore,

$$\boxed{\varphi(z) \approx 0.5720}.$$

Part (d): ReLU activation

The ReLU activation is

$$\varphi(z) = \max(0, z).$$

Since $z = 0.29 > 0$,

$$\varphi(0.29) = \max(0, 0.29) = 0.29.$$

Therefore,

$$\boxed{\varphi(z) = 0.29}.$$

Final Answer

Quantity	Value
Weighted sum z	0.29
Threshold output, $\theta = 1$	0
Logistic output	0.5720
ReLU output	0.29

Exam Notes

- Do not forget the bias term $w_0d_0 = (0.20)(1)$.
- The weighted sum z is computed once. Each activation function then transforms the same $z = 0.29$ in a different way.
- With threshold activation, compare z directly against the threshold θ . Here $0.29 < 1$, so the output is 0.
- With ReLU, positive values pass through unchanged and negative values are clipped to 0.